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Complex wireframe DNA origami nanostructures with multi-arm junction vertices

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Structural DNA nanotechnology[1–4] and the DNA origami technique[5], in particular, have provided a range of spatially addressable two- and three-dimensional nanostructures[6–10]. These structures are, however, typically formed of tightly packed parallel helices[5–9]. The development of wireframe structures[10,11] should allow the creation of novel designs with unique functionalities, but engineering complex wireframe architectures with arbitrarily designed connections between selected vertices in three-dimensional space remains a challenge. Here, we report a design strategy for fabricating finite-size wireframe DNA nanostructures with high complexity and programmability. In our approach, the vertices are represented by $n \times 4$ multi-arm junctions ($n = 2$–10) with controlled angles, and the lines are represented by antiparallel DNA crossover tiles[12] of variable lengths. Scaffold strands are used to integrate the vertices and lines into fully assembled structures displaying intricate architectures. To demonstrate the versatility of the technique, a series of two-dimensional designs including quasi-crystalline patterns and curvilinear arrays or variable curvatures, and three-dimensional designs including a complex snub cube and a reconfigurable Archimedean solid were constructed.

To construct an arbitrarily shaped wireframe architecture, the target pattern is treated as a planar graph[13], in that all the intersections between edges are vertices, and the two-dimensional (2D) combinatorial map[14] of the planar graph ensures the correct routing of the scaffold in the pattern. In the present design, the first step is to convert all the connections between vertices into double lines (Fig. 1a). The second step is to 'loop' and 'bridge' all the lines into a single continuity along which one single-stranded scaffold strand can travel through all the vertices. Proper crossovers between the lines are placed to make the scaffold strand go through all the lines unidirectionally, once and only once, so that in any individual line segment the two lines are antiparallel (Fig. 1a). Next, the complementary staple strands in the line segments are added following the rules for generating double-crossover (DX) DNA tiles[12], where antiparallel DX junctions between the lines are inserted to bridge the two DNA helices separated by a full number of DNA helical turns.

To adjust the angles between arms and assign the sequences, a T_n loop of appropriate length is inserted into the staple strands surrounding the vertex and a certain number of nucleotides in the scaffold strand are left unpaired opposite the T_n loop. The added unpaired nucleotides at the vertex provide a degree of structural flexibility and allow bending at the corners to generate the desired angles between the arms at the vertex (Fig. 1b) (see Supplementary Figs 3 and 38 for a detailed discussion). Additionally, the design of the staple strands requires consideration of the length of each of the line segments. The lengths of the line segments between any two neighbouring vertices need to be an integer number of DNA helical turns to minimize strain and to allow the scaffold DNA to follow the designed folding path. Figure 1c shows two possible scenarios when adding staple strands: with or without a scaffold crossover in the line segment (the latter requires a shift of the crossover positions of the staple strands to avoid juxtaposing two crossovers). This distributes the positions of the crossovers evenly along the line segments, which determines the integrity and rigidity of each part of the structure.

To demonstrate the design principles, we first constructed three Platonic tessellation patterns with homogeneous vertices and edges of regular polygonal patterns in the form of hexagonal, square or triangle geometries. A honeycomb pattern containing 16 hexagons is used as an example to illustrate the detailed design process. In this pattern, each vertex is joined by three arms with equal 120° angles. The target pattern is first laid out and each arm converted into two parallel lines (Supplementary Fig. 1). Each pair of lines meeting at the vertex are then connected and looped at the edges. This results in a number of individual hexagonal loops in the centre part and a continuous loop along the perimeter of the whole pattern (Supplementary Fig. 1). Next, DXs are inserted between the neighbouring hexagon loops at selected positions so that the individual loops are all connected to form a single loop. A final bridge is added between the internal and edge loops (Supplementary Fig. 2). Accordingly, one single-stranded circular DNA acting as a scaffold can go through the entire structure and visit each vertex and line three and two times, respectively.

The next step is to add the complementary staple strands in the line segments according to the design rules of forming DX tiles. Here, we used five full turns of B-form DNA as the length of each edge. Unpaired nucleotides around the vertices are manually assigned to satisfy the angle requirement as reported before[11,15–17]. Because the honeycomb array has a homogeneous 120° intersection angle, a T_3 loop in the staple strands at each corner is suitable to maintain this angle[15] and all the bases in the scaffold strand remain base-paired (Supplementary Fig. 3). The edges are treated differently depending on whether or not they contain a scaffold crossover (see Supplementary Fig. 4 for design details). Finally, nick points are added at carefully selected positions to obtain appropriate-length staple strands.

To generate the honeycomb structure (Fig. 2a), 7,100 of 7,249 nucleotides (nt) of M13mp18 single-stranded DNA (ssDNA) were assigned, and 223 staple strands with lengths ranging from 20 to 52 bases were used. The extra 138 nt of the scaffold is left as an unpaired loop on the outer edge. All the staple strands on the outermost edges were modified with poly T tails (Supplementary Fig. 4) to prevent potential π–π stacking along blunt-ended helices between individual structures.

Similar design principles were applied to create the other two Platonic patterns. The square tiling has 90° intersection angles, and four T_4 loops were used in the centre of each vertex to create the 90° angles[16] (Supplementary Fig. 11). The triangular tiling has 60° intersection angles, and six T_5 loops were used at each

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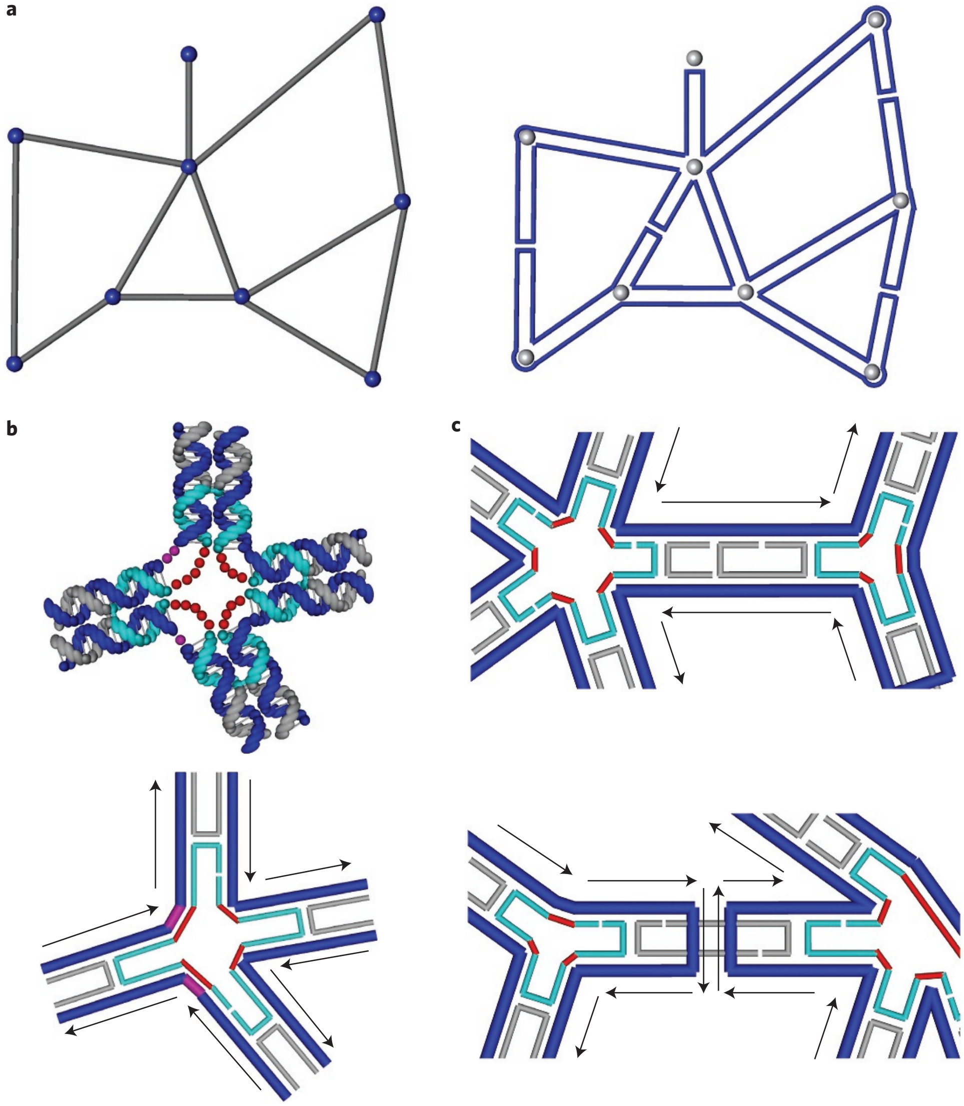

Figure 1 | Design principles. a, Left: an arbitrary wireframe pattern composed of line segments (grey) and vertices (blue). Right: steps to route a scaffold. First, double all the line segments from the original pattern. Second, connect the lines that meet at each vertex. Third, 'loop' and 'bridge' all the lines into one continuous scaffold. **b**, A DNA helical model of a 4 × 4 junction (top) and a line model of the 4 × 4 junction (bottom). Each vertex is designed as an $n \times 4$ junction. The angle of adjacent arms in one junction can be adjusted by inserting poly T loops (red dots) and leaving unpaired nucleotides in the scaffold strand opposite to the poly T loop (pink dots). **c**, Adding staple strands on two different types of edge (five-turn long edges are used here for illustration): the edge with two antiparallel scaffolds (top) and the edge with a scaffold bridge (Holliday junction) in the middle (bottom). Arrows indicate the direction from the 5′ end to the 3′ end of the DNA. In **b** and **c**, dark blue strands represent the scaffold strand and grey and cyan strands are the staple strands.

vertex[17] (Supplementary Fig. 11). The dimensions of the final pattern depend on the length of the scaffold strand available and the length of the edges (a minimum of three helical turns). By design, the dimensions of the 4 × 4 hexagonal pattern are 150 nm × 150 nm (the width of a DX tile is ~4.5 nm), the 5 × 5 square pattern measures 108 nm × 108 nm, and the triangular pattern is 136 nm × 136 nm in the 2D plane. Atomic force microscopy (AFM) images confirmed the successful formation of the designed structures (Fig. 2a–c) with high yield and structural integrity (see Supplementary Table 1 for yield analysis).

The present strategy can be easily adapted to the design of larger and more complex structures based on multiple scaffolds. For example, we designed a square tiling pattern using both M13mp18 (7,249 nt) and PhiX174 (5,386 nt) as the scaffold strands to create a 2D grid containing 8 × 9 cross-shaped vertices (or 7 × 8 square cavities). To ensure structural integrity, the contacts between the two scaffolds are maximized via staple strands. The final structure has four helical turns in each edge and provides a fully addressable square lattice of 148 nm × 167 nm (Fig. 2d). To further scale up the square lattice pattern, we took inspiration from our previously reported symmetric tile hierarchical assembly method[18] to assemble a square lattice containing 17 × 17 square cavities measuring ~300 nm in both dimensions (Fig. 2e).

Using this design strategy, we were able to create more complex wireframe structures without local translational symmetry (as shown in Fig. 1a) using quasi-crystalline structures as the target patterns. We demonstrated three patterns without translational symmetry. These include a star shape, a five-fold Penrose tiling and an eight-fold quasi-crystalline pattern (Fig. 3a–c). In contrast to the homogeneous tiling patterns, these three structures have the following features: (1) a diverse number of arms meeting at each vertex, ranging from three to eight arms; (2) various intersection angles between any neighbouring arms, ranging from 30° to 150°; and (3) variable arm lengths, from three to six turns. Successful

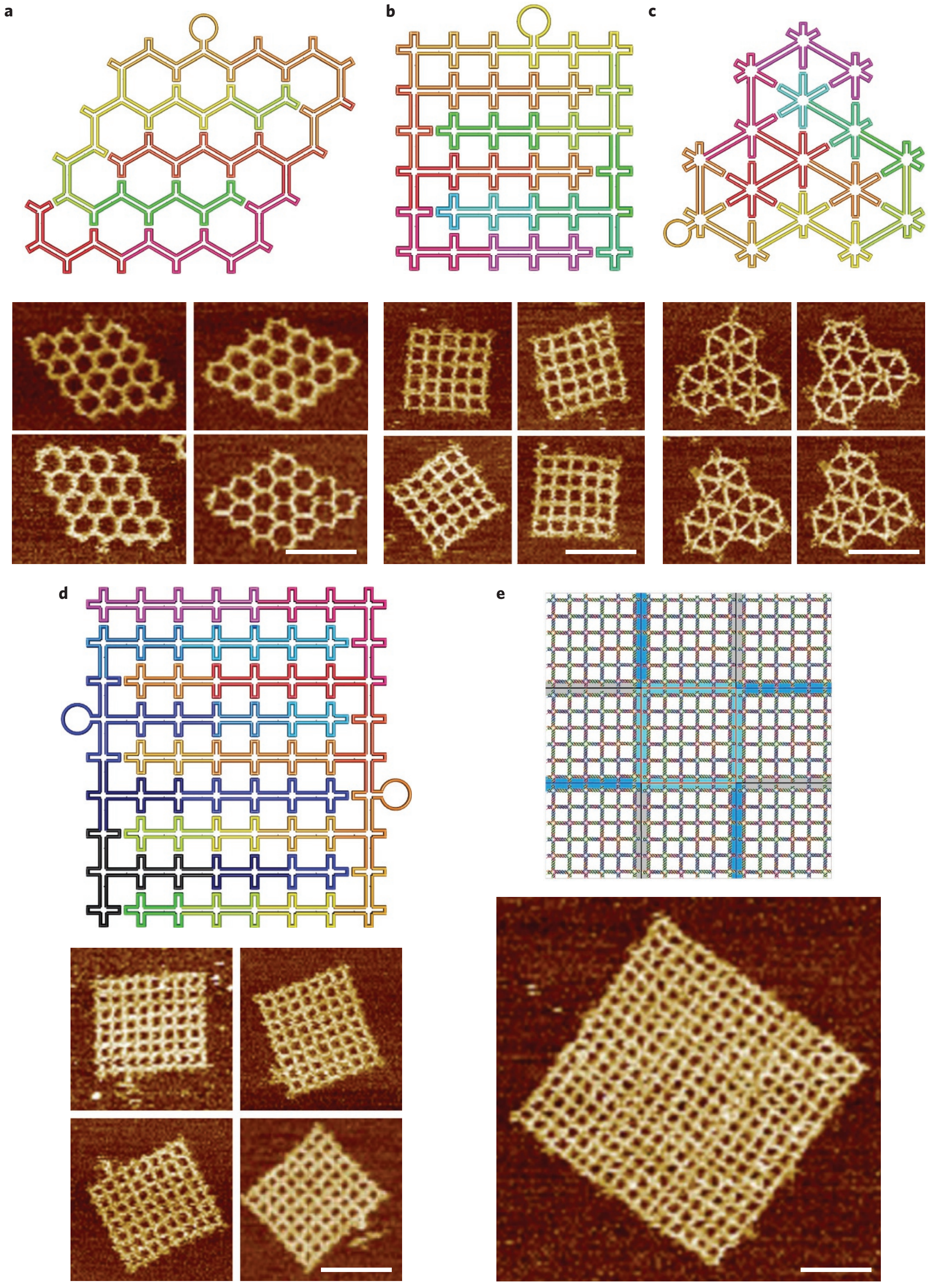

Figure 2 | Scaffold folding paths and representative AFM images for the simple Platonic tiling. a–c, Platonic tiling based on hexagon, square and triangle geometries, respectively. **d**, A 7 × 8 square Platonic tiling composed of two scaffolds (the black-blue loop on the left is a PhiX174 scaffold, and the coloured loop on the right is an M13mp18 scaffold). **e**, A lattice with 17 × 19 square cavities. All scale bars, 100 nm.

formation of these structures demonstrates the high programmability of our methods.

Various intersection angles are achieved by the insertion of T_n loops in the staple strands around each vertex and leaving unpaired bases on the scaffold strand between arms, as necessary (see Supplementary Figs 22–24 for design details). For example, in the Penrose tiling (Fig. 3b), surrounding the central five-arm vertex there are 5 three-arm vertices that have different angles between the three arms: 108°, 108° and 144° (Supplementary Fig. 23). T_3, T_3 and T_4 are inserted in the staple strand surrounding the vertex,

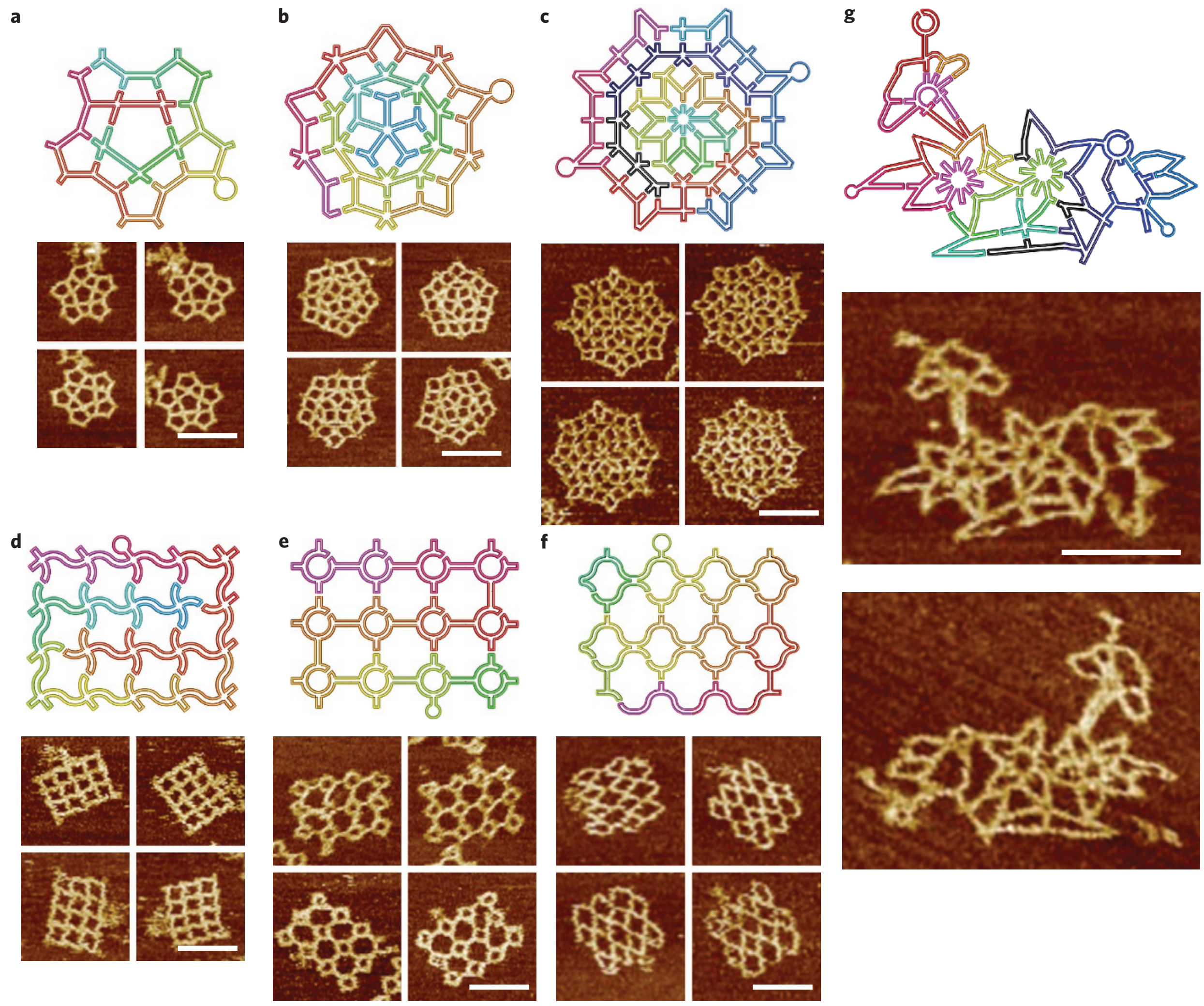

Figure 3 | Scaffold folding path and representative AFM images for intricate 2D patterns. a, A star-shaped pattern without translational symmetry. **b**, A Penrose tiling. **c**, An eight-fold quasi-crystalline pattern. **d–f**, Three curved structures: a wavy grid (**d**), a sphere array (**e**) and a fishnet (**f**). **g**, Flower-and-bird pattern. All scale bars, 100 nm. Two scaffolds (one colourful and one black-blue) are used in **c** and **g**.

and one unpaired nucleotide is left on the scaffold strand to achieve the 144° angle (Supplementary Fig. 23-2). The next group of vertices connected to the central one all contain five arms, with two 36°, one 72° and two 108° angles. These angles are achieved by inserting T_5 or T_4 loops into the staple strands and a T_5 loop opposite to an unpaired nucleotide on the scaffold (Supplementary Fig. 23-4). The number of added unpaired nucleotide(s) in the scaffold strand opposite the loop on the staple strand is chosen based on the distance between the neighbouring arms (see Supplementary Fig. 38 for a detailed discussion). The rules for creating different angles are summarized in Supplementary Fig. 38.

For the two quasi-crystalline patterns shown in Fig. 3, all the edges are of the same length. However, for the star-shaped pattern in Fig. 3a, the edges have three different lengths with a ratio of 6:5:2.9, which can be achieved by using six, five and three full helical turns in the arms, respectively. Therefore, various target structures with reduced local symmetry can be successfully synthesized by adjusting the angles and arm lengths. Structural characterization by native gel electrophoresis (Supplementary Fig. 46) and AFM imaging confirmed the successful assembly of the structures with high yield (Fig. 3a–c; see Supplementary Table 1 for yield analysis).

In the present design, every line segment of the patterns can be considered an analogue of a DX DNA tile (two double-helical DNA linked by two or three DXs). This feature enables us to introduce more intricate curvatures at each line segment to generate wavy or circular patterns (Fig. 3d–f), which can be achieved using a previously reported targeted insertion and deletion method[8]. Taking the wavy grid structure as an example (Fig. 3d), we laid down the scaffold strand on each line segment as two concentric arcs instead of two parallel lines. In each unit, a ~72° arc is created when a 31 base pair (bp) and a 21 bp double-helical DNA are laid side by side and connected by two crossovers at both ends (see Supplementary Figs 25 and 26 for details). By rotating and attaching such segments together, waving patterns of arbitrary curvatures can be generated (Supplementary Figs 25–34).

To test the design methods for producing patterns of extreme complexity, a more intricate and arbitrary shape in the form of three flowers and a bird was constructed by means of angle variation and curvature creation, as discussed already (Fig. 3g). This structure consists of vertices with from two to ten arms, a wide range of different angles between the arms, and various arm lengths and curvatures. We folded two scaffold strand loops and individually modified each of the vertices and lines to create the desired angles and local curvatures (see Supplementary Figs 35–37 for details) with an assembly yield of 73% (Supplementary Table 1). AFM imaging clearly shows the correct formation of the structures to resemble the designed patterns, demonstrating the versatility of

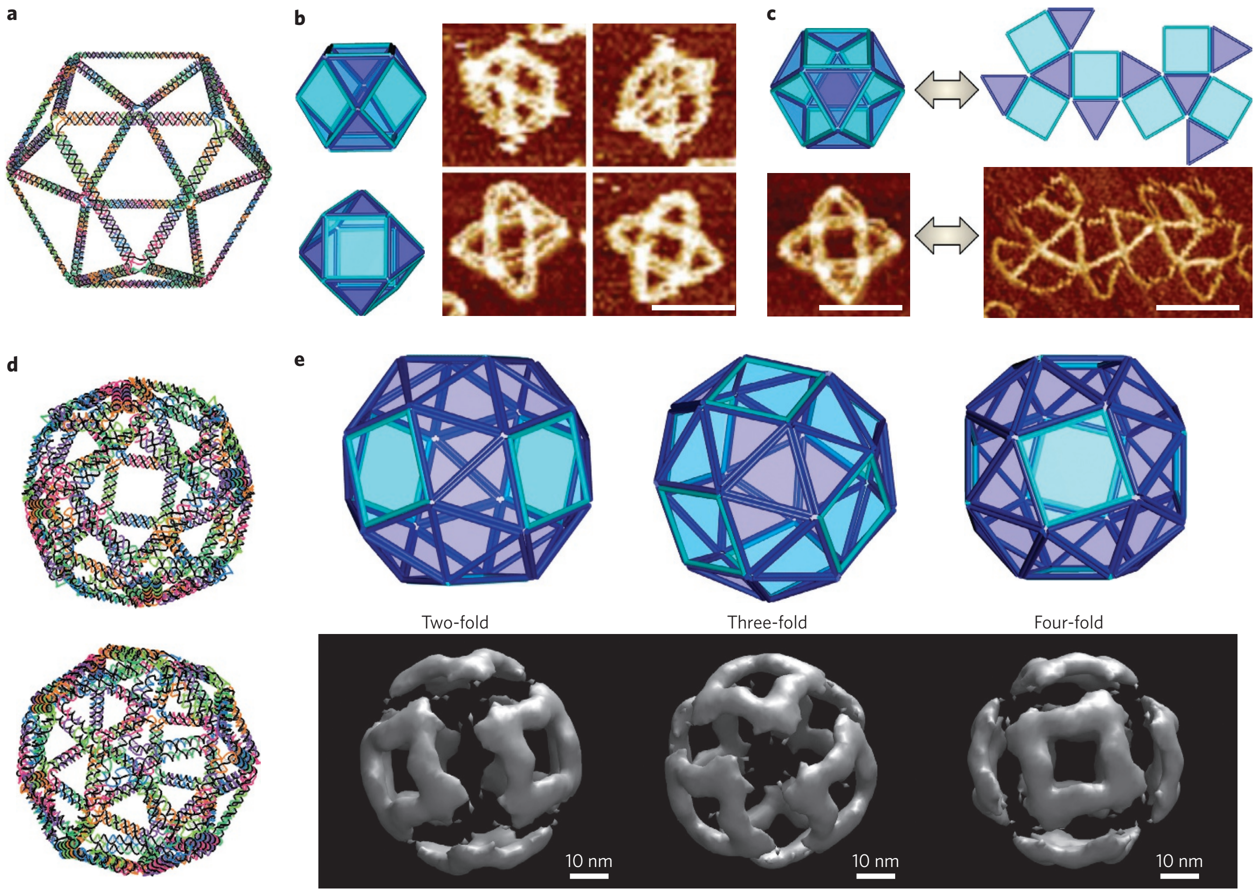

Figure 4 | 3D wireframe Archimedean solid structures. a, A 3D model of an Archimedean solid cuboctahedron with 12 vertices and 24 edges. Each vertex is a 4 × 4 junction and each edge is a 14-turn-long double DNA duplex. **b**, Left: models showing possible conformations of the structure when deposited on a mica surface. Right: corresponding AFM images. **c**, Reconfiguration between three and two dimensions can be realized by strand displacement by adding fuel and set strands. Top: reconfiguration schematics. Bottom: AFM images showing the transition. All scale bars in AFM images, 100 nm. **d**, 3D model of a snub cube with 24 vertices and 60 edges. Each vertex is a 5 × 4 junction and each edge is a five-turn double DNA duplex. **e**, Three views of the DNA snub cube from the design model (top) and density maps reconstructed from cryo-EM images (bottom) along the two-, three- and four-fold rotational symmetry axes, respectively.

our strategy for creating very sophisticated wireframe DNA nanostructures, which have been difficult to achieve using previous methods.

The design principle can be adapted easily to generate three-dimensional (3D) architectures with vertices and edges arranged in 3D space with or without structural symmetry. This can be achieved by following the same design steps, from turning single line edges into two lines, 'looping' and 'bridging,' adjusting angles and filling in staples. One important step for 3D construction is identifying appropriate folding paths and bridge points to fold the scaffold into a single loop that winds through all parts of the structure. A simple solution is to use the topological equivalent 2D polygon net (Schlegel diagram[19]) for the 3D structures to allow the use of the 'loop' and 'bridge' strategy demonstrated in two dimensions to generate the looping path of the scaffold[20–22] (Supplementary Figs 42 and 43). Angles in each vertex are adjusted using the same method by inserting T_n loops in the staples around the vertices and leaving unpaired nucleotides in the scaffold at the vertices (Supplementary Figs 41 and 44).

As shown in Fig. 4, a simple Archimedean solid cuboctahedron was designed and constructed to contain 12 equivalent four-arm vertices (with 60°, 90°, 60° and 90° angles) and equal-length two-helix DNA as edges, each 14 full DNA helical turns long. The structure can be easily observed under AFM (Fig. 4b,c) in a flattened form when it is deposited on a mica surface. The majority of the objects observed are centred with one of the square surfaces, indicating that the square surface has a higher tendency (stronger binding affinity) to contact the mica surface than the triangular surfaces or any vertex. Flattening of the 3D shape on the surface in this preferred way involves only angle changes at the vertices and minimal stretching or compression of any edges, which is expected as the angle change at the single-stranded loops around the vertices should involve much smaller free energy costs than stretching or compressing the DNA helices.

Reconfigurable transitions between the 2D and 3D conformations were achieved using the strand displacement technique[23]. On adding corresponding fuel and set strands, the staples on the outer edges of the 2D net were released and replaced by new staples that joined the specific edges together to form the 3D conformation. The new staples on the 3D polyhedron were also designed with a new set of toehold strands to allow the unfolding transformation from three to two dimensions (see Supplementary Figs 79 and 80 for details). The unfolding–refolding reconfigurations were clearly verified by AFM imaging (Fig. 4c and Supplementary Figs 63 and 64), providing further proof for correct 3D structure formation.

To demonstrate the complexity of 3D wireframe DNA structure design, we constructed a snub cube object, which is an Archimedean solid with 60 edges, 24 vertices and 38 faces including 6 squares and 32 equilateral triangles. Each edge was designed as a five-turn-long

DNA double helix. The five-arm vertices were assigned as T_5, T_5, T_5, T_5 and T_4 loops between adjacent arms, corresponding to 60°, 60°, 60°, 60° and 90° angles. Cryogenic transmission electronic microscopy (cryo-EM) provided direct evidence for the formation of the snub cube (Fig. 4e and Supplementary Fig. 65). For experimental details see Methods. The cryo-EM study confirmed the overall geometry, and all key elements showed up in the reconstructed model. The observed edges are ~20 nm long and 6 nm thick, which closely matches the design parameters (22 nm long and 5 nm thick). The two-, three- and four-fold symmetries in the snub cube were confirmed by a cryo-EM structural model (Fig. 4e). The triangular faces are geometrically rigid, but the square faces are deformable and would introduce some structural flexibility into the DNA snub cube. The cryo-EM study confirmed such an expectation, as some electron densities surrounding the square faces were missing in the reconstructed structural model. The 3D structures shown here each may have two possible conformational diastereomers[24] (that is, one of these structures turned inside out has the same connectivity), which cannot be distinguished due to the current low resolution of our cryo-EM imaging. To obtain the information of handedness from cryo-EM, the resolution should be high enough to distinguish the major and minor grooves of the DNA helix[24]. In the future, the conformational diastereomers may be rationally controlled by introducing rigid stapling joints at selected corners of the structure.

In summary, we have demonstrated the utility of a set of new design rules for engineering wireframe DNA nanostructures with unprecedented complexity. In principle, this strategy can be applied to design and construct any imaginable wireframe nanostructure. In the examples demonstrated here we used antiparallel DX junctions to form 'bridges' between two neighbouring helices. Other DNA or RNA junction motifs, such as paranemic crossovers (PX)[25] and RNA kissing loops[26], can also be used to enrich the design and produce interesting topological structures that could potentially be replicated *in vivo* by biological machinery. The structural platform reported here enables the construction of nanostructures with a new level of structural complexity, thus creating new opportunities for engineering functional materials. For example, the fabrication of quasi-crystalline patterns with rational design and programmability has been a challenging task. The method described here offers a new way to program the formation of quasi-crystalline patterns for the templated assembly of functional nanomaterials with emergent properties. Compared with traditional DNA origami structures of tightly packed parallel helices, the structures demonstrated here would allow easy incorporation of mechanical springs and/or tensegrity properties into the addressable edges to construct responsive nanomechnical devices and 3D porous nanomaterials.

Methods

Methods and any associated references are available in the online version of the paper.

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Author contributions

H.Y., Y.L. and F.Z. conceived and designed the experiment. F.Z., S.J., S.W. and Y.L. performed the experiments. F.Z., S.J., S.W. and Y.L. analysed the data. All authors discussed the results. All authors contributed to the writing the manuscript.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to Y.L. and H.Y.

Competing financial interests

The authors declare no competing financial interests.

Methods

Materials. All DNA staple strands were purchased in 96-well plates from Integrated DNA Technologies (www.IDTDNA.com) at 25 nmol synthesis scales with concentrations normalized to 200 μM. Staple strands were used directly from plates without further purification. Scaffold single-stranded M13mp18 viral DNA and phiX174 DNA were purchased from New England Biolabs (catalogue numbers N4040S and N3023S).

Individual wireframe DNA origami nanostructure assembly. Structures with a single scaffold were assembled by mixing 5 nM single-stranded M13mp18 DNA (7,249 nucleotides) with a tenfold molar excess of staple strands in 1× TAE Mg^{2+} buffer (40 mM Tris, 20 mM acetic acid, 2 mM EDTA and 12.5 mM magnesium acetate [pH 8.0]). Frame origami structures with two scaffolds were annealed by mixing 5 nM M13mp18 and 5 nM phiX174 with the staple strands in a 1:1:20 molar ratio in 1× TAE Mg^{2+} buffer. The final volume of the mixture was 100 μl. The design details and sequences of the DNA oligos used to form each structure are provided in Supplementary Section 5. The resulting solutions were annealed in a polymerase chain reaction thermocycler from 90 °C to 4 °C in ~12 h: from 90 °C to 86 °C at a rate of 4 °C per 5 min, from 85 °C to 70 °C at a rate of 1 °C per 5 min, from 70 °C to 40 °C at a rate of 1 °C per 15 min, from 40 °C to 25 °C at 1 °C per 10 min, then held at 4 °C at the end of the cycle.

Hierarchical assembly of 3 × 3 square lattice origami array. The three-square lattice origami with different extended sticky-end staple strands were individually annealed by following the protocol described above. Excess staple strands were washed away using a 100 kDa MWCO (molecular weight cut-off) Amicon centrifugal filter (Sigma-Aldrich). A 100 μl annealed sample and 400 μl 1× TAE-Mg^{2+} buffer were added into the filter. After spinning for 3 min at 2,850 r.c.f. (relative centrifugal force), another 400 μl 1× TAE-Mg^{2+} buffer was added for the second wash. After three washes, the three structures with the appropriate stoichiometry (1:4:4) were mixed together and left at room temperature (25 °C) for 12 h before characterization.

Transformation of the cuboctahedron between the 2D open net and 3D shape. The starting structure either in two or three dimensions was annealed using the protocol described above. Without further purification, twofold molar excesses of fuel strands were added to the sample and incubated at 45 °C for 6 h. A four times molar excess of set strands were then added to the sample and cooled from 45 °C to 4 °C at a rate of 1 °C per 10 min.

AFM imaging. AFM imaging was conducted in the 'ScanAsyst mode in fluid' (Dimension FastScan, Bruker) with Scanasyst-Fluid+tips (Bruker). Sample were prepared for AFM imaging as described in the following. A 2 μl annealed sample was deposited onto a freshly cleaved mica surface (Ted Pella) and 3 μl $NiCl_2$ (25 mM) was added immediately to the samples (the effect of adding $NiCl_2$ is discussed in Supplementary Section 3). After waiting ~30 s for adsorption to the mica surface, 80 μl 1× TAE-Mg^{2+} buffer was added to the sample, and an extra 40 μl of the same buffer was deposited on the AFM tip.

Cryo-EM imaging. The snub cube concentration was 5 nM after annealing. The sample was concentrated with 100 K membrane tubes (Millipore Amicon Ultra, Ultracel 100 K Membrane) from 1,700 μl to 45 μl. A 3 μl concentrated solution (particle concentration of 125 nM) was deposited on a Quantifoil R 2/2 grid. The grid was blotted with two pieces of filter paper (Whatman, 1001-055) and immediately plunge-frozen using liquid nitrogen-cooled liquid ethane. Images were taken on a Titan Krios TEM with an accelerating voltage of 300 kV under low-dose conditions to minimize radiation damage to the samples. To enhance the image contrast, an under-focus in the range of 7–9 μm was used to record the images using a Gatan US4000 4k × 4k charge-coupled device camera.

Single-particle reconstruction. Computer software EMAN2[27] was used for single-particle reconstruction. In total, 500 particles were selected for the reconstruction process with the EMAN2 reconstruction package following the standard protocol. All data were processed following the EMAN2 reconstruction tutorial using the workflow interface (http://blake.bcm.edu). The following general steps were applied: import and pre-process micrographs; picking particles; CTF determination; remove bad particles and build particle sets; make 2D reference free class-averages; build an initial 3D model; 3D high-resolution refinement; evaluate resolution of a 3D map. To generate initial models, 156 particles were selected. The initial orientations of individual particles were randomly assigned within the corresponding asymmetric units of the octahedron symmetry for the particle. In total, 384 particles were used for 3D refinements of the snub cube. The refinement was carried out with a 2° angle interval. A projection-matching algorithm was applied to determine the centre and orientation of the raw particles in the iterative refinement for the prism. The resulting 3D map was generated after seven iterations of refinement, when it reached convergence. The symmetry was then relaxed to C1 (no symmetry) and refined for another four iterations with a 3° angle interval to confirm the featured symmetry (C4, C3, C2) of the particle. After refinement, class averages of the raw particles and 2D projections of the 3D model were generated in two files, but paired in order. The comparisons presented were manually selected to show the distinct and comprehensive orientations. The resolution of the resulting structural density map was determined to be at 8.5 nm, with Fourier shell correlation (0.5 threshold criterion). The final reconstruction results were visualized with UCSF Chimera[28] software.

Native agarose gel electrophoresis. Native gel electrophoresis was conducted on 0.5% agarose gel in 1× TAE-Mg^{2+} buffer at 80 V sitting in an ice-water bath. Next, 5 μl annealed sample was mixed with 1 μl 100× SYBR Gold nucleic acid gel stain solution (Life Technologies) and the mixture was loaded into the cast gel well. After running for 2–3 h, the gel was visualized under ultraviolet light (Gel Doc XR+ system gel imager, Bio-Rad). The yield analysis based on agarose gel is presented in the Supplementary Information.

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